Jacobson is Not a [P] Substitute: What 1995 Proves, and What 2016 Does Not

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Abstract

This note contains only **[P]** statements.

Jacobson 1995, under the hypotheses he states—entropy proportional to area, $\delta Q = T\,dS$, Unruh temperature, heat as boost-energy flux, local Rindler horizons instantaneously stationary, and $\nabla^a T_{ab} = 0$—implies the Einstein equation with an *undetermined* cosmological constant and with no torsion [**?**]. The null-quadratic lemma in that argument is an identity and is re-derived here. **[P]**

Jacobson 2016 states that for conformal fields the maximal vacuum entanglement hypothesis is equivalent to Einstein at first order, and that for nonconformal fields the same conclusion holds *modulo a conjecture* about $\delta\langle K\rangle$ [**?**]. The installed Standard Model is not a CFT ($b_3 = -7$). The conformal half therefore does not apply to NSM matter. The nonconformal half is not **[P]**, by Jacobson's own words. **[P]**

Paper IX's closure bar for cosmology is FGHMV-grade Einstein–Cartan including Hehl–Datta. Jacobson 1995 produces Einstein with free Λ and no spin. Jacobson 2016 does not become **[P]** for the SM. Therefore Jacobson is not a **[P]** substitute for Q2.

What is allowed on this page

A statement is written only if it is a logical implication, an identity, a citation of a sentence in Jacobson, or a number already computed from the Standard Model representation table. No **[C]**. No dual. No promotion of MVEH.

1 The 1995 implication

Jacobson assumes $dS = \eta\,\delta\mathcal{A}$, $T = \hbar\kappa/2\pi$, $\delta Q = \int T_{ab}\chi^a\,d\Sigma^b$ on a local Rindler horizon with $\theta = \sigma = 0$ at the bifurcation surface, and $\nabla^a T_{ab} = 0$. Raychaudhuri then gives $\theta = -\lambda R_{ab}k^a k^b$ near the surface, and $\delta Q = T\,dS$ forces

$$T_{ab}k^a k^b = \frac{\hbar\eta}{2\pi}\,R_{ab}k^a k^b \qquad \text{for every null } k. \tag{1}$$

The algebraic content of (**??**) is the following lemma.

Lemma 1.1 (null quadratic form). *Let S_{ab} be symmetric on Lorentzian $\mathbb{R}^{1,3}$. If $S_{ab}k^a k^b = 0$ for every null k, then $S_{ab} = f\,\eta_{ab}$.* ***[P]***

Proof. Mostly minus, $k = (1, n)$ with $n \cdot n = 1$. Then $S_{00} + 2S_{0i}n^i + S_{ij}n^in^j = 0$ for every unit n. The odd part forces $S_{0i} = 0$. Polarization on $\{e_i, (e_i + e_j)/\sqrt{2}\}$ is a six-by-seven linear system whose kernel is one-dimensional and is spanned by $S_{00} = -\lambda$, $S_{ij} = \lambda\delta_{ij}$. That is $S = \lambda\eta$. The auditor solves that system; the kernel is exactly $(-1, 1, 1, 1, 0, 0, 0)$ when S_{00} is normalized to -1.

Theorem 1.2 (Jacobson 1995, algebra isolated)**.** *Lemma* **??** *and* $\nabla^a T_{ab} = 0$ *together with the contracted Bianchi identity convert* (**??**) *into*

$$R_{ab} - \tfrac{1}{2}R\, g_{ab} + \Lambda g_{ab} = \frac{2\pi}{\hbar\eta}\, T_{ab}, \qquad G = \frac{1}{4\hbar\eta}, \tag{2}$$

with Λ *an arbitrary constant.* ***[P]***

Jacobson writes that Λ "remains as enigmatic as ever" [**?**]. The constant is not fixed by $\delta Q = T\, dS$. If $\nabla^a T_{ab} = 0$ is dropped, f in $T_{ab} = \alpha(R_{ab} + f g_{ab})$ need not be $-R/2 + \Lambda$. That mutation is why conservation is load-bearing.

Theorem 1.3 (no Einstein–Cartan, no Hehl–Datta)**.** *The 1995 input list is* (g, R_{ab}, T_{ab}, k)*. It does not contain an independent connection, a spin tensor, or* J_5*. The output* (**??**) *is Einstein, not Einstein–Cartan, and it does not contain* $\mathcal{L}_{\mathrm{HD}} = -(3\kappa/16)J_5 \cdot J_5$*.* ***[P]***

2 The 2016 split, in Jacobson's words

Jacobson 2016, abstract and §IV: for first-order conformal fields the Einstein equation holds if and only if the maximal vacuum entanglement hypothesis holds; "for nonconformal fields, the same conclusion follows modulo a conjecture about the variation of entanglement entropy." He then writes, verbatim, that he will "conjecture—and we shall assume" that

$$\delta\langle K\rangle = \frac{\Omega_{d-2}\ell^d}{d^2 - 1}\, \delta\langle T_{00}\rangle + \delta X \tag{3}$$

when the field is not conformal. He also states that finiteness and universality of the UV area density η "remains an assumption."

Those sentences are the paper. They are not upgraded here.

The conformal half uses $K = H_\zeta = \int T^{ab}\zeta_b\, d\Sigma_a$, the Casini–Huerta–Myers operator of a CFT. The installed Standard Model has one-loop $b_3 = -7 \neq 0$ (Dynkin count, auditor CONFIRMED). It is not a CFT. The conformal half of 2016 does not apply to NSM matter. **[P]**

3 Why this is not a [P] substitute for Q2

Paper IX closes cosmology only if one has (i) a family of de Sitter regions with controlled modular Hamiltonians, (ii) a first-law identity, (iii) equivalence to linearized Einstein–Cartan including the Cartan equation, (iv) Hehl–Datta from relative-entropy positivity.

Theorem 3.1 (not a substitute)**.** *Jacobson 1995 gives (Einstein,* Λ *free) from Clausius and an area law, and gives none of (iii), (iv). Jacobson 2016 is not* ***[P]*** *for the Standard Model. Therefore neither paper meets the Paper IX bar. Jacobson is not a* ***[P]*** *substitute for Q2.* ***[P]***

Local Rindler horizons exist at every point of de Sitter. That geometric fact is why 1995 can be *stated* in dS. It does not turn (**??**) into a derivation of Λ, of Einstein–Cartan, or of a holographic pair.

4 Machine check

Solver `m9_7_jacobson.py`; auditor `m9_7_audit_jacobson.py`. Lemma **??** is the kernel computation. $b_3 = -7$ is the species sum. Both PASS.

5 Verdict

Claim	Status
Lemma **??**	[**P**], audited
1995 $\Rightarrow$ Einstein, Λ free, $G = 1/(4\hbar\eta)$	[**P**] (Jacobson; algebra isolated)
1995 produces HD or EC	**false**, Thm. **??**
2016 conformal half applies to the SM	**false** ($b_3 \neq 0$)
2016 nonconformal half	not [**P**] (Jacobson's conjecture)
η finite and universal	assumption (Jacobson)
Jacobson is a [**P**] substitute for Q2	**false**, Thm. **??**

Admission. I did not derive Einstein–Cartan in de Sitter from Jacobson. I did not promote MVEH. I recorded the 1995 implication under its hypotheses and Jacobson's own restriction of 2016. That is all that is [**P**].

References

[1] T. Jacobson, Phys. Rev. Lett. 75 (1995) 1260, arXiv:gr-qc/9504004.

[2] T. Jacobson, Phys. Rev. Lett. 116 (2016) 201101, arXiv:1505.04753.